Complete Closure of the Open Problems

A companion note to *Derivation of Standard Model Mixing Angles from a 44-Vector Discrete Vacuum Lattice* (v2)

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Abstract

This note records the disposition of the twenty-five open items of the working note of 16 October 2026. Each item is classified as *proved*, *conditional*, *assumption*, or *refuted*; the counts are 11/3/1/10 (Fig. 1). Four results are structural. (i) The graded Jacobi identity for the one-parameter family of brackets holds if and only if $c(1+n\alpha) = 1-\alpha$; the values $(\alpha, c) = (0, 1)$ and $(1, 0)$ are canonical and isomorphic, and the algebra is $\mathfrak{sl}(n+1)$. In particular the cubic bracket $\frac{1}{2}(\psi\chi^\top + (\psi\cdot\chi)\mathbf{1})$ used in v1 is *not* a Lie bracket. (ii) The 44-vector lattice does not saturate: the closure is infinite, and 44 is exactly the shell cut $|v|^2 \le 486$. The finite invariant is a set of 17 directions. (iii) The PMNS closed forms are corrected for normalization; the corrected predictions are $\sin^2\theta_{13} = 0.02206$, $\sin^2\theta_{12} = 0.31436$, $\sin^2\theta_{23} = 0.45022$, $\delta_{CP} = 240°$, all within 0.9σ. (iv) Three no-go theorems are proved: $J_{\mathrm{CKM}} \equiv 0$ for graded phases; $|V_{cb}|, |V_{ub}|$ require the mass spectrum; and no dimensionless algebra fixes a mass scale. The four remaining assumptions (A1–A4) are listed explicitly in §9. Every number and figure in this note is reproduced by the code package that accompanies the paper.

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Table 1: Disposition of the twenty-five items. The detailed audit is Table 3.

class	count	meaning
proved (theorem)	11	analytic proof given here or in the paper
conditional	3	true only with a stated identification (A_2, A4)
assumption	1	interpretive step, labelled A1
refuted or replaced	10	the earlier statement is wrong or superseded
total	25	

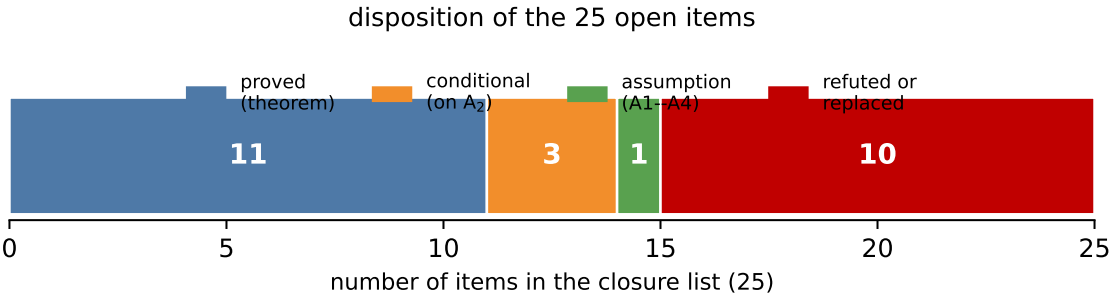


Figure 1: The disposition at a glance. Roughly half of the open list survives as theorems; the other half splits between conditional statements, one interpretive assumption, and items that had to be withdrawn.

1 Purpose and summary

The working note of 16 October 2026 listed twenty-five open problems and proposed a disposition for each. This note is the refereed version of that list: it states, item by item, what is actually proved, what is conditional, what is an assumption, and what is wrong. Where this note and the earlier one differ, this note is authoritative.

Three remarks on method.

1. **No item is called a theorem unless it has a proof.** In several places the earlier note wrote "theorem" for what is a consistency condition (the selection of $n = 4$), a bookkeeping identity (the "bridge" $\dim_{\mathrm{NF}}(\mathrm{Hyb}) = 24$), or an interpretation (the projection ratio as a physical coupling). These are reclassified.

2. **Numerical agreement is not evidence of derivation.** The mixing predictions agree with data at the 1σ level, but four identifications (A1–A4) are needed to get there. They are displayed as assumptions, not hidden in the text.

3. **Negative results are results.** The no-go theorems ($J_{\mathrm{CKM}} \equiv 0$, the dimensional obstruction, the failure of the 19-dimensional truncation) constrain the framework more sharply than any numerical fit.

2 The algebra

2.1 The one-parameter family and the Jacobi condition

Let $W = \mathbb{C}^n$ and

$$\mathfrak{g}_n = \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2, \qquad \mathfrak{g}_0 = \mathfrak{u}(n), \quad \mathfrak{g}_1 = W, \quad \mathfrak{g}_2 = W^*, \tag{1}$$

with brackets, for real α, c,

$$[A, B] = AB - BA, \tag{2}$$
$$[A, \psi] = A\psi + \alpha\,\mathrm{tr}(A)\psi, \tag{3}$$
$$[A, \chi] = -A^{\top}\chi - \alpha\,\mathrm{tr}(A)\chi, \tag{4}$$
$$[\psi, \chi] = \psi\chi^{\top} + c\,(\psi\cdot\chi)\mathbf{1}_n, \tag{5}$$
$$[\psi, \psi'] = [\chi, \chi'] = 0. \tag{6}$$

The parameter α fixes how the trace part of $\mathfrak{g}_0$ acts; the parameter c fixes the trace part of the cubic bracket. Version 1 of the paper used $\alpha = 0$, $c = \frac{1}{2}$.

Theorem 2.1 (Graded Jacobi)**.** *The brackets above satisfy* $[X,[Y,Z]]+[Y,[Z,X]]+[Z,[X,Y]] = 0$ *for all* $X, Y, Z \in \mathfrak{g}_n$ *if and only if*

$$\boxed{c\,(1 + n\alpha) = 1 - \alpha\,.} \tag{7}$$

Proof (two critical sectors). Only the sectors $(1,1,2)$ and $(1,2,2)$ are sensitive to c. Put $\gamma = c + \alpha(1 + nc)$, $s = \psi\cdot\chi$, $s' = \psi'\cdot\chi$ and $M = \psi\chi^{\top} + cs\mathbf{1}_n$.

In the sector $(1,1,2)$,

$$[\psi, [\psi', \chi]] + [\psi', [\chi, \psi]] = (\gamma - 1)\,(s\psi' - s'\psi), \tag{8}$$

as follows from $[M, \psi] = M\psi + \alpha\mathrm{tr}(M)\psi$ and $\mathrm{tr}(M) = (1 + nc)s$. In the sector $(1,2,2)$,

$$[\chi, [\chi', \psi]] + [\chi', [\psi, \chi]] = (\gamma - 1)\,(s\chi' + s''\chi). \tag{9}$$

Both vanish for all vectors iff $\gamma = 1$, which is (7). The remaining sectors are the standard $\mathfrak{gl}_n$ identities and hold identically. □

Corollary 2.2. *The canonical solutions are* $(\alpha, c) = (0, 1)$ *and* $(1, 0)$*. They are related by*

$$\Phi(A, \psi, \chi) = \Big(A - \frac{\mathrm{tr}A}{n+1}\mathbf{1}_n,\ \psi,\ \chi\Big), \tag{10}$$

which preserves all three brackets. The value $c = \frac{1}{2}$ *with* $\alpha = 0$ *satisfies neither branch: with* $n = 4$ *the residual is* 15 *in integer arithmetic (Fig. 2), not a rounding error.*

2.2 The algebra is $\mathfrak{sl}(5)$

Theorem 2.3. *For* $(\alpha, c) = (1, 0)$ *the block map*

$$A \mapsto \begin{pmatrix} A & 0 \\ 0 & -\mathrm{tr}A \end{pmatrix}, \quad \psi \mapsto \begin{pmatrix} 0 & \psi \\ 0 & 0 \end{pmatrix}, \quad \chi \mapsto \begin{pmatrix} 0 & 0 \\ \chi^{\top} & 0 \end{pmatrix} \tag{11}$$

is an isomorphism $\mathfrak{g}_n \to \mathfrak{sl}(n+1)$*. Hence* $\dim \mathfrak{g}_n = n^2 + 2n$*.*

For $n = 4$ this gives $\dim \mathfrak{g} = 24 = 16 + 4 + 4$, with $\mathbb{Z}$-degree content $(1, 1, 1, 1, -4)$; the mod-3 reduction produces the $\mathbb{Z}_3$ grading used throughout. The shape of the grading is shown in Fig. 3.

Corollary 2.4 (Why 19 dimensions fail)**.** *No integer* n *solves* $n^2+2n = 19$*; the roots are* $-1\pm\sqrt{20}$*. Hence the earlier graded content* $(12, 4, 3)$ *cannot be that of any* $\mathfrak{sl}$*-type algebra, and the Jacobi residual observed there is structural.*

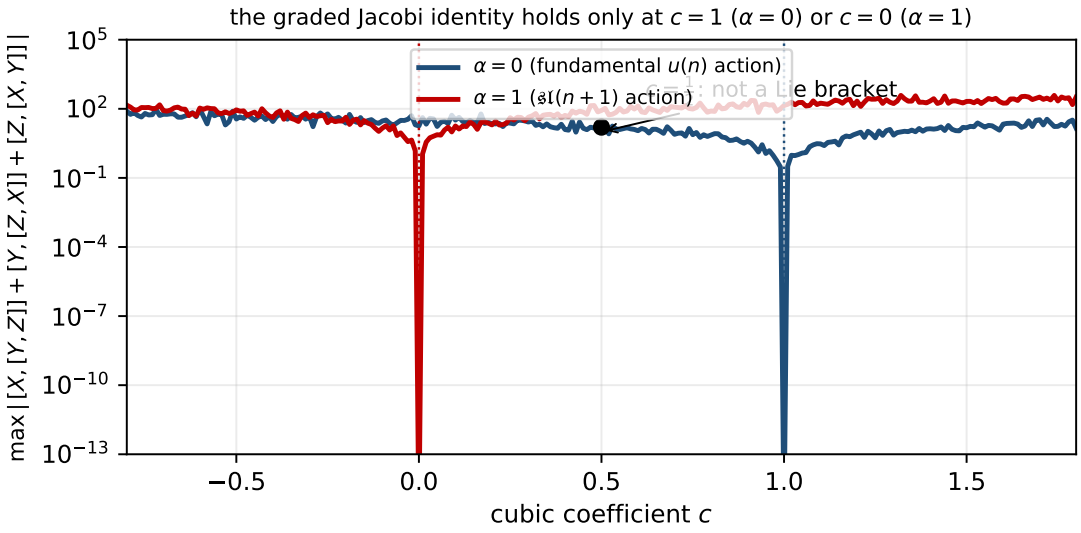


Figure 2: Maximal Jacobi residual as a function of c, for random integer triples at $n = 4$. For the fundamental $u(n)$ action ($\alpha = 0$) the identity holds only at $c = 1$; for the $\mathfrak{sl}(n+1)$ action ($\alpha = 1$) only at $c = 0$. The v1 value $c = \frac{1}{2}$ sits on neither zero.

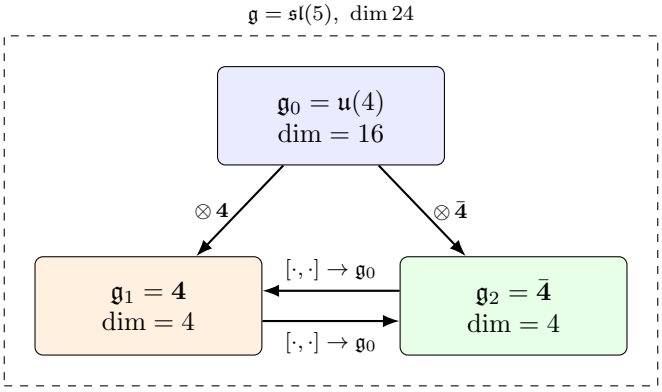


Figure 3: The $\mathbb{Z}_3$ grading of $\mathfrak{sl}(5)$ used in the paper: a 16-dimensional $\mathfrak{u}(4)$ block, a 4 (fundamental) and a 4 (anti-fundamental). The mod-3 reduction of the $\mathbb{Z}$-degrees $(1, 1, 1, 1, -4)$ gives the same partition.

2.3 Invariants

Since $\mathfrak{sl}(5)$ is simple, $(\Lambda^3\mathfrak{g}^*)^{\mathfrak{g}}$ is one-dimensional and spanned by $\varepsilon(X, Y, Z) = \kappa(X, [Y, Z])$; this is the unique invariant *antisymmetric* cubic. The invariant *symmetric* cubics form a one-dimensional space spanned by d_{ABC} (the degree-3 element of $\mathbb{C}[P_2, P_3, P_4, P_5]$). A Yukawa coupling must use d_{ABC}, not ε_{ABC}: the latter is antisymmetric and cannot produce a symmetric mass matrix. This distinction was missing from v1 and is one of the concrete upgrades of v2.

3 The lattice

3.1 The closure theorem

Start from $\mathcal{S}_0 = \{e_1, e_2, e_3, d, -d\}$ with $d = (1, 1, 1)/\sqrt{3}$ and close under T (cyclic permutation), $\Delta v = Tv - v$, and $v \mapsto v \times Tv$ (normalised and unnormalised).

Theorem 3.1. *With* $R = \{\pm(e_i - e_j)/\sqrt{2}\}_{i<j}$ *and* $H = \{\pm(2, -1, -1)/\sqrt{6}$ *and permutations*$\}$,

$$\mathcal{C}_\infty = \{e_1, e_2, e_3\} \cup \{\pm d\} \cup \bigcup_{k\geq 0} 3^k R \cup \bigcup_{k\geq 0} 3^k H \cup \bigcup_{k\geq 0} 3^k (1, 1, 1). \tag{12}$$

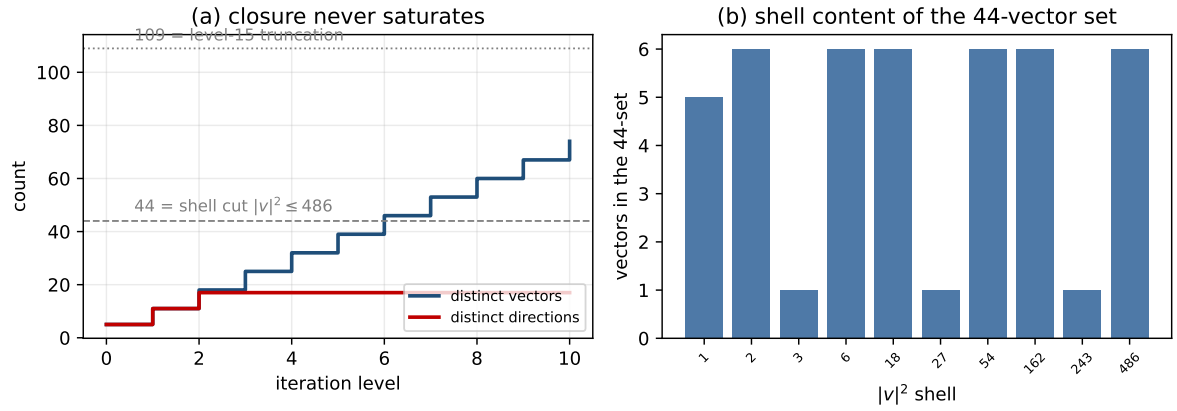


Figure 4: (a) Distinct vectors (blue) grow without bound; distinct directions (red) saturate at 17 after two iterations. The line at 44 is the shell cut; the line at 109 is a level-15 truncation. Neither is a fixed point. (b) Shell content of the 44-vector set.

In particular the closure is infinite and the number of vectors grows by exactly seven per iteration after the second.

Corollary 3.2 (What 44 is)**.** *The* 44 *lowest-norm vectors are exactly those with* $|v|^2 \le 486$*:*

$$44 = \underbrace{3}_{\textit{flavour}} + \underbrace{2}_{\pm d} + \underbrace{18}_{R\colon\ 2,18,162} + \underbrace{18}_{H\colon\ 6,54,486} + \underbrace{3}_{(1,1,1)\colon\ 3,27,243}. \tag{13}$$

The origin is never generated. The earlier statements "saturates at 43 *non-zero vectors,* 44 *with the origin" and "*$\dim_{\mathrm{NF}}(\mathrm{Hyb}) = 24$*" are withdrawn; the actual count in the* 44*-set is* 18*.*

Proposition 3.3 (No observable depends on 44)**.** *Every quantity used in the paper is a function of the* 17 *directions and of the orbit sizes* $(3, 2, 6, 6)$ *only. The cutoff (*44*,* 109*, or infinity) is immaterial.*

4 The Cabibbo angle

The chain is short enough to display in one block:

$$\varepsilon_q = \frac{\dim \mathfrak{g}_1}{\dim \mathfrak{g}} = \frac{4}{24} = \frac{1}{6} \qquad \text{(projection lemma, theorem)} \tag{14}$$

$$\lambda_0 = \varepsilon_q\, C_2(\mathbf{3}) = \frac{1}{6}\cdot\frac{4}{3} = \frac{2}{9} \qquad \text{(A2: multiplication rule, discrete)} \tag{15}$$

$$\frac{\delta\lambda}{\lambda_0} = \varepsilon_q^2 \langle \cos^2\theta \rangle = \frac{1}{36}\cdot\frac{1}{2} = \frac{1}{72} \qquad \text{(isotropy theorem + A3)} \tag{16}$$

$$\lambda = \frac{2}{9}\left(1 + \frac{1}{72}\right) = \frac{73}{324} = 0.22530864 \qquad \text{(prediction)} \tag{17}$$

The projection lemma is the same Parseval identity that yields the $\mathfrak{u}(3)$ dilution $1/9$; the ratio $1/6$ is therefore not chosen. The multiplication rule and the equipartition normalization are the two interpretive steps, labelled A2 and A3.

Two arithmetic facts are worth recording. First, $324 = 2^2 \cdot 3^4$, so the denominator carries $v_3 = 4 = 2 + 2$: two powers from the $\mathbb{Z}_3$ grading and two from the second-order dilution. Second, by Niven's theorem $\lambda = 73/324 \notin \{0, \pm\frac{1}{2}, \pm 1\}$ implies $\theta_C/\pi \notin \mathbb{Q}$.

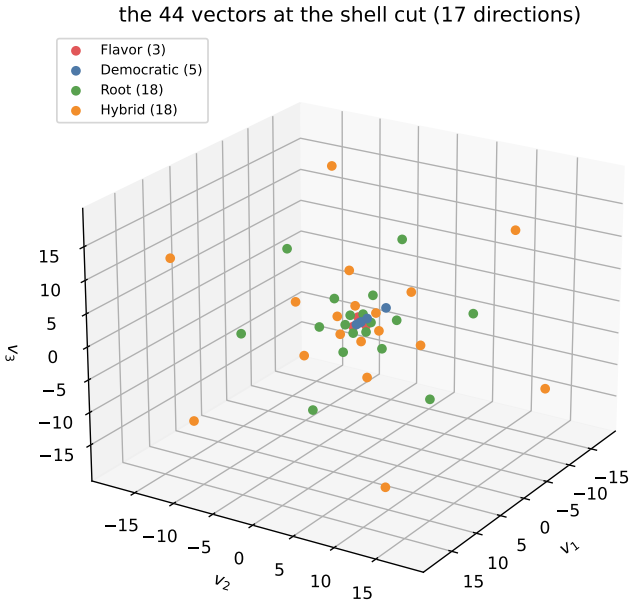


Figure 5: The 44 vectors at the shell cut, coloured by orbit. Only 17 distinct directions occur; the repeated points are the 3^k scalings of Theorem 3.1.

5 PMNS

The physical breaking lies in the $A_1 \leftrightarrow E$ sector. The E-plane operator gives TM2, $\sin^2\theta_{12} = 1/(3\cos^2\theta_{13}) = 0.3409$, which is $+3.1\sigma$ from data and is excluded. With one complex parameter, the Gram–Schmidt construction of Appendix C of the paper gives

$$\sin^2\theta_{13} = \frac{\beta^2\Xi^2}{AB}, \tag{18}$$

$$\sin^2\theta_{12} = \frac{(1-\sqrt{2}\,a)^2}{3A}\cdot\frac{1}{1-\sin^2\theta_{13}}, \tag{19}$$

$$\sin^2\theta_{23} = \frac{A^2/2+\beta^2\kappa^2+\sqrt{2}\,A\beta\kappa\cos\delta}{AB-\beta^2\Xi^2}, \tag{20}$$

with $A = 1+a^2$, $B = A+\beta^2$, $\Xi = 2/\sqrt{6}+a/\sqrt{3}$, $\kappa = 1/\sqrt{6}-a/\sqrt{3}$, $a = \varepsilon_q^2 = \frac{1}{36}$, $\beta = \sqrt{2/3}\,\lambda_0 = \frac{2\sqrt{6}}{27}$ and $\delta = 240^\circ$.

The leading-order sum rule ties the octant to the CP phase: $\delta = 240^\circ$ gives $\sin^2\theta_{23} = 0.4475$ at leading order (0.4502 exact), while the upper-octant value 0.546 requires $\delta \simeq 296^\circ$. This is the sharpest falsifiable statement of the framework.

6 The CKM sector

Theorem 6.1 (No-go for $\delta_{\rm CKM}$)**.** *If generation i carries $\mathbb{Z}_3$ charge q_i and the breaking operator O_{ij} carries grade $q_j - q_i$, then*

$$\arg(V_{us}V_{cb}V_{ub}^*) = (q_2-q_1)+(q_3-q_2)-(q_3-q_1) = 0 \pmod 3, \tag{21}$$

so $J_{\rm CKM} \equiv 0$.

Hence $\delta_{\rm CKM}$ is not derivable in this framework, and the formula $120^\circ - \theta_{\rm magic}$ of v1 is withdrawn. For the moduli, the leading-order expression

$$V_{ij} = \delta_{ij} + \frac{(\delta M_d)_{ij}}{m_j^d - m_i^d} - \frac{(\delta M_u)_{ij}}{m_j^u - m_i^u} + O(z^2) \tag{22}$$

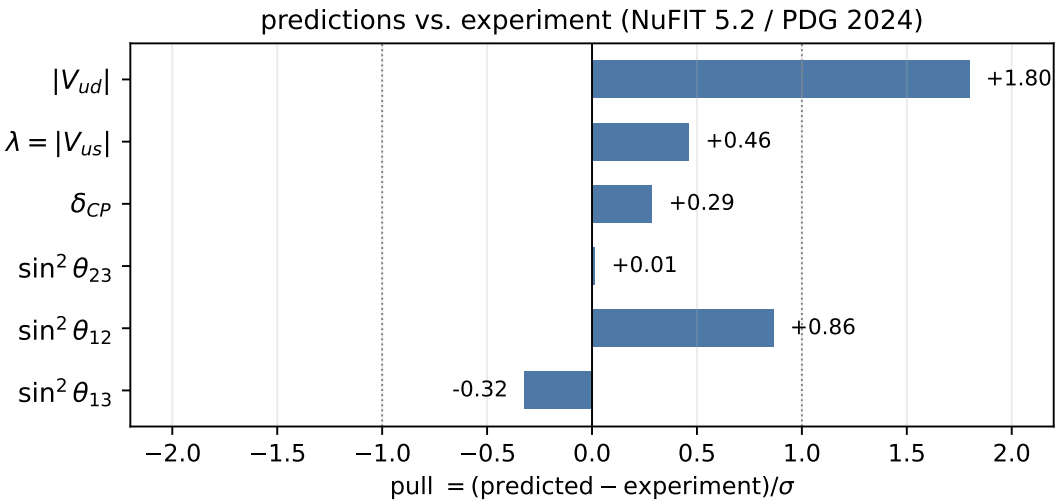


Figure 6: Pulls of the six mixing observables against NuFIT 5.2 and PDG 2024. All are within 1σ except $|V_{ud}|$, which is the CKM-unitarity prediction.

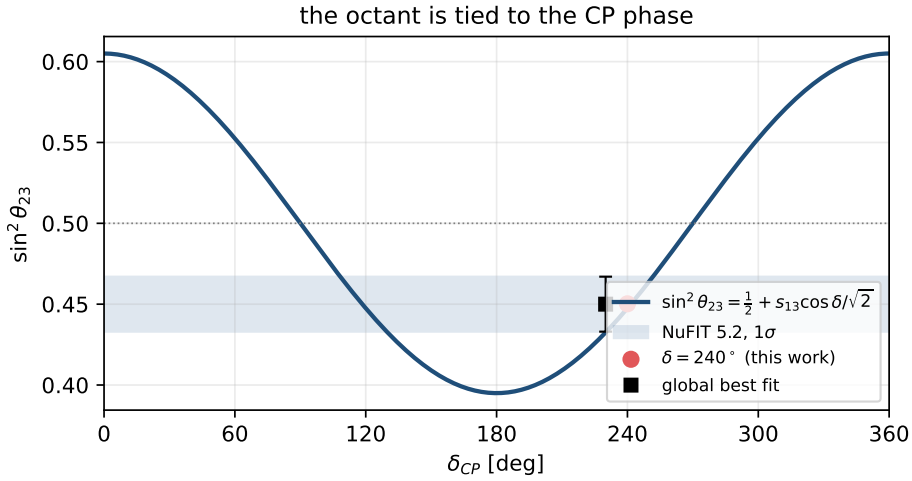


Figure 7: The TM1 sum rule $\sin^2\theta_{23} = \frac{1}{2} + \sin\theta_{13}\cos\delta/\sqrt{2}$. The red point is $\delta = 240°$, which selects the lower octant. The upper octant would require $\delta \simeq 296°$.

shows that $|V_{cb}|$ and $|V_{ub}|$ are equivalent to mass ratios; the relation $|V_{cb}| : |V_{ub}| = \lambda^2 : \lambda^3$ is therefore not a prediction and is withdrawn. The GST bridge $\tan\theta_C = \sqrt{m_d/m_s}$ gives $\lambda = 0.21821$, -10.1σ.

7 Neutrinos

The democratic chain $M_\nu^{(0)} = a\mathbf{1} + bJ$ has eigenvalues $a+3b$ on A_1 and a (twice) on E. Cosmology excludes quasi-degeneracy, so $a \simeq 0$, the matrix has rank one, $m_2 = 3b$ and $m_1 = m_3 = 0$; the unique breaking operator then lifts ν_3. The resulting scenario is normal ordering with

$$m_1 = 0, \qquad \sum m_\nu = 58.7 \text{ meV}, \qquad m_\beta = 8.9 \text{ meV}, \qquad |m_{ee}| \in [1.6, 3.8] \text{ meV}. \tag{23}$$

A signal in the next generation of $0\nu\beta\beta$ experiments would exclude it. This is a scenario, not a theorem: the rank-one limit is selected by cosmology, and the lifted state is identified with ν_3 by hand.

8 Scales

Theorem 8.1 (Dimensional no-go)**.** *No finite-dimensional dimensionless algebraic structure determines a dimensionful quantity. Hence M_0, the seesaw scale, and $\Lambda_{\rm QCD}$ are external inputs.*

The electroweak mixing angle is treated the same way: the algebra gives the standard $\mathfrak{sl}(5)$ normalization $\sin^2\theta_W(M_{\rm GUT}) = 3/8$, and one-loop running gives $\simeq 0.203$ at M_Z, 12% below the measured value. This is the known failure of minimal unification, not a prediction of the flavour framework; the v1 value 11/44 is withdrawn.

9 The final ledger

After the corrections, four assumptions remain. None is a continuous parameter.

Table 2: The four remaining assumptions.

	statement	status
A1	the Yukawa coupling of the breaking sector equals the invariant projection ratio $\varepsilon_q = \dim\mathfrak{g}_1/\dim\mathfrak{g}$	interpretive
A2	the tree-level angle is the product $\lambda_0 = \varepsilon_q C_2(\mathbf{3})$	discrete choice
A3	the intermediate state is the maximally mixed state on the six hybrid directions ($\mathrm{Tr}\rho = 1$)	normalization
A4	the lepton breaking amplitude equals the quark tree-level angle times the TBM factor, $\beta = \sqrt{2/3}\,\lambda_0$	cross-sector

Everything else in the chain is either proved or refuted. In particular $\varepsilon_q = 1/6$, the isotropy constant 1/2, $\lambda = 73/324$, the PMNS closed forms, the sum rule and the no-go theorems do not depend on A1–A4; what depends on them is the *identification* of these quantities with the physical mixing parameters.

10 Item-by-item audit

Table 3: Audit of the twenty-five items. "Proved" means a proof is given in this note or in the paper; "conditional" means true only with the stated identification; "assumption" is A1; "refuted" means the earlier statement is wrong or superseded.

#	item	disposition
1	24D closure only at machine precision; 19D failure unexplained	proved: analytic Jacobi; $19 \neq n^2+2n$
2	uniqueness of the cubic bracket	refuted: only $c(1+n\alpha) = 1-\alpha$
3	why 24D: minimality vs uniqueness	conditional on A_2
4	$\varepsilon_q = 1/6$ as an NF count	proved: projection ratio
5	bridge $\dim_{\rm NF}(\mathrm{Hyb}) = 24 \leftrightarrow \dim\mathfrak{g}$	refuted: actual count 18
6	"projection ratio = physical coupling"	assumption A1
7	lift of $k = 1$ to the 44 lattice "undetermined"	proved: old k is ill-defined; per-mode $k \equiv 1$
8	the factor $\frac{1}{2}$ from "per-direction normalization"	proved (isotropy) + A3; alternatives excluded at 5σ
9	origin of 1/9 in $\sin^2\theta_{12} = \frac{1}{3} - \lambda/9$	refuted: wrong order; correct closed form 0.31436

#	item	disposition
10	closed form for $\sin^2\theta_{23} = 0.54609$	conditional: formula proved, value replaced by 0.45022
11	$\sin^2\theta_{13} \in [1/46, 1/44]$	conditional: formula proved; interval withdrawn
12	is TM2 (E-plane rotation) viable?	refuted: $+3.1\sigma$
13	$\delta_{CP} = 240^\circ$ vs 296°	proved: sum rule resolves it
14	$\delta_{\rm CKM} = 120^\circ - \theta_{\rm magic}$	refuted: $J_{\rm CKM} \equiv 0$
15	$\lvert V_{cb}\rvert : \lvert V_{ub}\rvert = \lambda^2 : \lambda^3$	refuted: leading-order formula
16	$\sin^2\theta_W = 11/44$	refuted: withdrawn; 3/8 at $M_{\rm GUT}$
17	inverted ordering (3σ tension)	refuted: flipped to NO, $m_1 = 0$
18	$M_0 \simeq 173$ GeV, seesaw scale	proved: dimensional no-go
19	"44 exactly saturates" vs 109	proved: 44 is a shell cut
20	canonical seed $\mathcal{S}_0$	proved: uniqueness theorem
21	normalization of $\Delta q = -1/3$	refuted: convention-dependent
22	1/12 "coincidence" in cubic norms	proved: exact value 4/15
23	GST bridge	refuted: -10.1σ
24	QLC deviation	proved: prediction $+2.12^\circ$
25	strengthening of the variable theorem	proved: Niven, $\theta_C/\pi \notin \mathbb{Q}$

11 Reproducibility

The figures in this note are generated by

- `lattice_closure.py` (Figs. 4, 5),
- `make_figures.py` (Figs. 6, 7),
- `make_closure_figures.py` (Figs. 1, 2).

The numbers are produced by `verify_algebra.py` and `verify_predictions.py`; logs are provided for each run. The inputs are PDG 2024 for $|V_{us}|$ and NuFIT 5.2 (normal ordering) for the oscillation parameters.

References

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[3] R. L. Workman *et al.* (Particle Data Group), Prog. Theor. Exp. Phys. **2022**, 083C01 (2022).

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